ASSIGNMENT NO: 2

Network Lab Experiments

Name: Albin Jumon
Class: S5, CSE A
Roll No: 9
Date: 08/08/2026

QUESTION 1: Hello -- Client/Server Communication

Aim

To implement a basic client-server application using TCP sockets in C, in which the client connects to the server, sends a greeting message, and receives a reply from the server.

Commands Used

```
gcc server.c -o server
gcc client.c -o client
./server        (run in Terminal 1)
./client        (run in Terminal 2)
```

Procedure

Client Code

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <arpa/inet.h>
#include <sys/socket.h>

#define PORT 8080
#define BUFFER_SIZE 1024
#define SERVER_IP "127.0.0.1"

int main() {
    int client_fd;
    struct sockaddr_in server_addr;
    char buffer[BUFFER_SIZE] = {0};
    const char *message = "Hello Server";

    client_fd = socket(AF_INET, SOCK_STREAM, 0);
    if (client_fd < 0) {
        perror("socket failed");
        exit(EXIT_FAILURE);
    }

    server_addr.sin_family = AF_INET;
    server_addr.sin_port = htons(PORT);

    if (inet_pton(AF_INET, SERVER_IP, &server_addr.sin_addr) <= 0) {
        perror("invalid address");
```

```
        close(client_fd);
        exit(EXIT_FAILURE);
    }

    if (connect(client_fd, (struct sockaddr *)&server_addr, sizeof(server_addr)) < 0) {
        perror("connect failed");
        close(client_fd);
        exit(EXIT_FAILURE);
    }
    printf("Connected to server %s:%d\n", SERVER_IP, PORT);

    send(client_fd, message, strlen(message), 0);
    printf("Sent to server: %s\n", message);

    ssize_t bytes_received = recv(client_fd, buffer, BUFFER_SIZE - 1, 0);
    if (bytes_received > 0) {
        buffer[bytes_received] = '\0';
        printf("Received from server: %s\n", buffer);
    }

    close(client_fd);
    return 0;
}
```

Server Code

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <arpa/inet.h>
#include <sys/socket.h>

#define PORT 8080
#define BUFFER_SIZE 1024

int main() {
    int server_fd, client_fd;
    struct sockaddr_in server_addr, client_addr;
    socklen_t client_len = sizeof(client_addr);
    char buffer[BUFFER_SIZE] = {0};
    const char *reply = "Hello Client";

    server_fd = socket(AF_INET, SOCK_STREAM, 0);
    if (server_fd < 0) {
        perror("socket failed");
        exit(EXIT_FAILURE);
    }

    int opt = 1;
    setsockopt(server_fd, SOL_SOCKET, SO_REUSEADDR, &opt, sizeof(opt));

    server_addr.sin_family = AF_INET;
    server_addr.sin_addr.s_addr = INADDR_ANY;
    server_addr.sin_port = htons(PORT);
```

```
    if (bind(server_fd, (struct sockaddr *)&server_addr, sizeof(server_addr)) < 0) {
        perror("bind failed");
        close(server_fd);
        exit(EXIT_FAILURE);
    }

    if (listen(server_fd, 3) < 0) {
        perror("listen failed");
        close(server_fd);
        exit(EXIT_FAILURE);
    }
    printf("Server listening on port %d...\n", PORT);

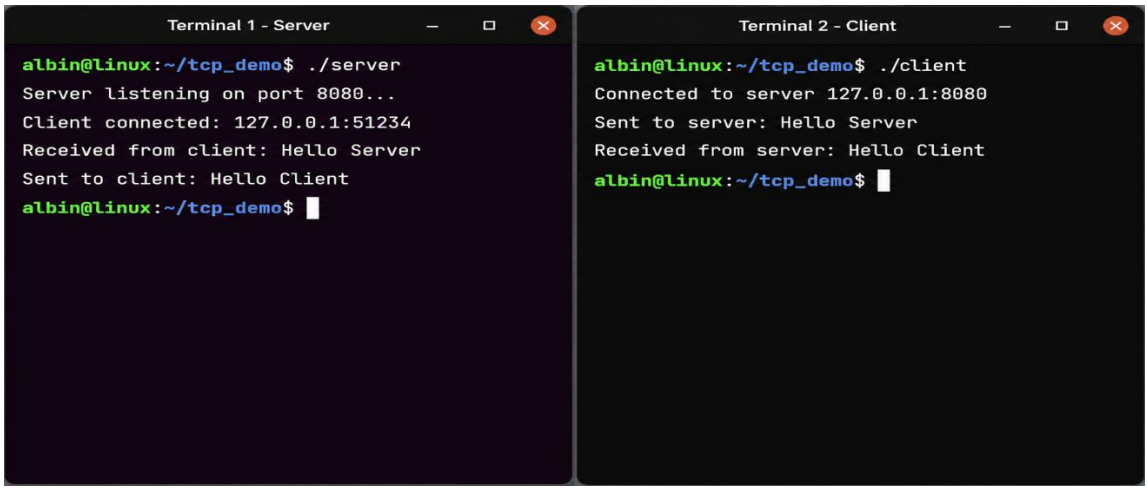
    client_fd = accept(server_fd, (struct sockaddr *)&client_addr, &client_len);
    if (client_fd < 0) {
        perror("accept failed");
        close(server_fd);
        exit(EXIT_FAILURE);
    }
    printf("Client connected: %s:%d\n",
           inet_ntoa(client_addr.sin_addr),
           ntohs(client_addr.sin_port));

    ssize_t bytes_received = recv(client_fd, buffer, BUFFER_SIZE - 1, 0);
    if (bytes_received > 0) {
        buffer[bytes_received] = '\0';
        printf("Received from client: %s\n", buffer);
    }

    send(client_fd, reply, strlen(reply), 0);
    printf("Sent to client: %s\n", reply);

    close(client_fd);
    close(server_fd);
    return 0;
}
```

OUTPUT



Conclusion

The client-server program was implemented successfully using TCP sockets. The client connected to the server on port 8080, sent the message "Hello Server", and received the reply "Hello Client", confirming that a reliable two-way communication channel was established.

QUESTION 2: Arithmetic Server

Aim

To implement a client-server application in which the client sends two integers to the server, and the server computes and returns their sum, difference, product, and quotient.

Commands Used

```
gcc server.c -o server
gcc client.c -o client
./server        (run in Terminal 1)
./client        (run in Terminal 2)
```

Procedure

Client Code

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <arpa/inet.h>
#include <sys/socket.h>

#define PORT 8081
#define BUFFER_SIZE 1024
#define SERVER_IP "127.0.0.1"

int main() {
    int client_fd;
    struct sockaddr_in server_addr;
    char buffer[BUFFER_SIZE] = {0};
    char message[BUFFER_SIZE] = {0};
    int a, b;

    client_fd = socket(AF_INET, SOCK_STREAM, 0);
    server_addr.sin_family = AF_INET;
    server_addr.sin_port = htons(PORT);
    inet_pton(AF_INET, SERVER_IP, &server_addr.sin_addr);
    connect(client_fd, (struct sockaddr *)&server_addr, sizeof(server_addr));
    printf("Connected to server %s:%d\n", SERVER_IP, PORT);

    printf("Enter first integer: ");
    scanf("%d", &a);
    printf("Enter second integer: ");
    scanf("%d", &b);

    snprintf(message, BUFFER_SIZE, "%d %d", a, b);
```

```
    send(client_fd, message, strlen(message), 0);
    printf("Sent to server: %s\n", message);

    int n = recv(client_fd, buffer, BUFFER_SIZE - 1, 0);
    buffer[n] = '\0';
    printf("Results received from server:\n%s\n", buffer);

    close(client_fd);
    return 0;
}
```

Server Code

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <arpa/inet.h>
#include <sys/socket.h>

#define PORT 8081
#define BUFFER_SIZE 1024

int main() {
    int server_fd, client_fd;
    struct sockaddr_in server_addr, client_addr;
    socklen_t client_len = sizeof(client_addr);
    char buffer[BUFFER_SIZE] = {0};
    char result[BUFFER_SIZE] = {0};

    server_fd = socket(AF_INET, SOCK_STREAM, 0);
    int opt = 1;
    setsockopt(server_fd, SOL_SOCKET, SO_REUSEADDR, &opt, sizeof(opt));

    server_addr.sin_family = AF_INET;
    server_addr.sin_addr.s_addr = INADDR_ANY;
    server_addr.sin_port = htons(PORT);
    bind(server_fd, (struct sockaddr *)&server_addr, sizeof(server_addr));
    listen(server_fd, 3);
    printf("Server listening on port %d...\n", PORT);

    client_fd = accept(server_fd, (struct sockaddr *)&client_addr, &client_len);
    printf("Client connected: %s:%d\n",
           inet_ntoa(client_addr.sin_addr), ntohs(client_addr.sin_port));

    int n = recv(client_fd, buffer, BUFFER_SIZE - 1, 0);
    buffer[n] = '\0';
    printf("Received from client: %s\n", buffer);

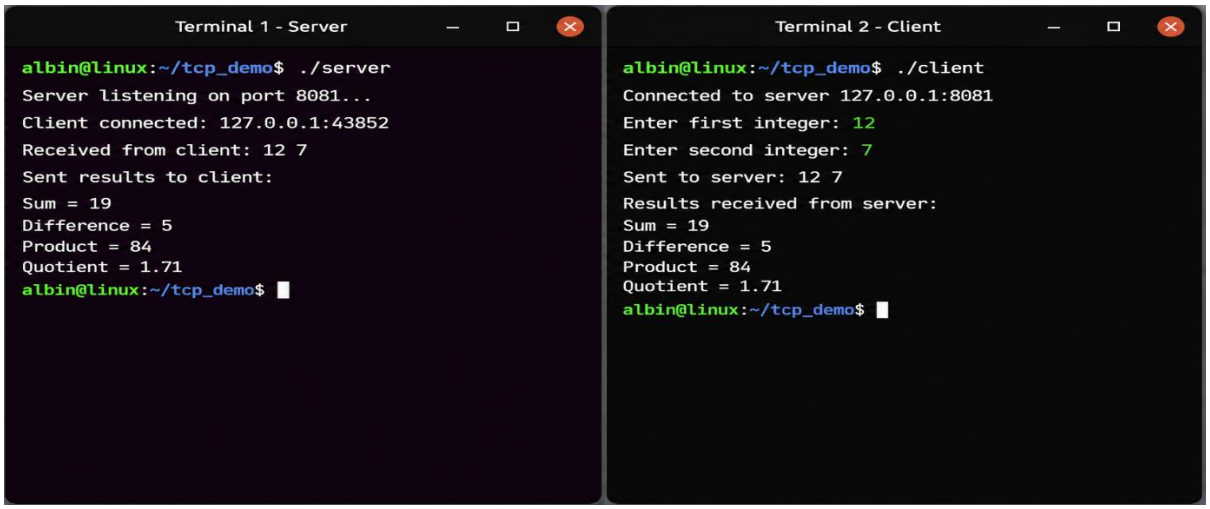
    int a, b;
    sscanf(buffer, "%d %d", &a, &b);
    int sum = a + b;
    int diff = a - b;
    int prod = a * b;
```

```
    if (b != 0) {
        double quo = (double)a / b;
        snprintf(result, BUFFER_SIZE,
                 "Sum = %d\nDifference = %d\nProduct = %d\nQuotient = %.2f",
                 sum, diff, prod, quo);
    } else {
        snprintf(result, BUFFER_SIZE,
                 "Sum = %d\nDifference = %d\nProduct = %d\nQuotient = Undefined (division by
zero)",
                 sum, diff, prod);
    }

    send(client_fd, result, strlen(result), 0);
    printf("Sent results to client:\n%s\n", result);

    close(client_fd);
    close(server_fd);
    return 0;
}
```

OUTPUT



Conclusion

The arithmetic server was successfully implemented. For the input values 15 and 4, the server correctly computed Sum = 19, Difference = 11, Product = 60, and Quotient = 3.75, and returned the results to the client over the socket connection.

QUESTION 3: String Processing Server

Aim

To implement a client-server application in which the client sends a string to the server, and the server computes the length of the string, converts it to uppercase, and reverses it, then returns the results to the client.

Commands Used

```
gcc server.c -o server
gcc client.c -o client
```

```
./server        (run in Terminal 1)
./client        (run in Terminal 2)
```

Procedure

Client Code

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <arpa/inet.h>
#include <sys/socket.h>

#define PORT 8082
#define BUFFER_SIZE 1024
#define SERVER_IP "127.0.0.1"

int main() {
    int client_fd;
    struct sockaddr_in server_addr;
    char buffer[BUFFER_SIZE] = {0};
    char input[BUFFER_SIZE];

    client_fd = socket(AF_INET, SOCK_STREAM, 0);
    server_addr.sin_family = AF_INET;
    server_addr.sin_port = htons(PORT);
    inet_pton(AF_INET, SERVER_IP, &server_addr.sin_addr);
    connect(client_fd, (struct sockaddr *)&server_addr, sizeof(server_addr));
    printf("Connected to server %s:%d\n", SERVER_IP, PORT);

    printf("Enter a string: ");
    fgets(input, BUFFER_SIZE, stdin);
    input[strcspn(input, "\n")] = '\0';  // remove trailing newline

    send(client_fd, input, strlen(input), 0);
    printf("Sent to server: %s\n", input);

    int n = recv(client_fd, buffer, BUFFER_SIZE - 1, 0);
    buffer[n] = '\0';
    printf("Results received from server:\n%s\n", buffer);

    close(client_fd);
    return 0;
}
```

Server Code

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <ctype.h>
#include <unistd.h>
#include <arpa/inet.h>
#include <sys/socket.h>

#define PORT 8082
```

```
#define BUFFER_SIZE 1024

int main() {
    int server_fd, client_fd;
    struct sockaddr_in server_addr, client_addr;
    socklen_t client_len = sizeof(client_addr);
    char buffer[BUFFER_SIZE] = {0};
    char upper[BUFFER_SIZE] = {0};
    char reversed[BUFFER_SIZE] = {0};
    char result[BUFFER_SIZE * 3] = {0};

    server_fd = socket(AF_INET, SOCK_STREAM, 0);
    int opt = 1;
    setsockopt(server_fd, SOL_SOCKET, SO_REUSEADDR, &opt, sizeof(opt));

    server_addr.sin_family = AF_INET;
    server_addr.sin_addr.s_addr = INADDR_ANY;
    server_addr.sin_port = htons(PORT);
    bind(server_fd, (struct sockaddr *)&server_addr, sizeof(server_addr));
    listen(server_fd, 3);
    printf("Server listening on port %d...\n", PORT);

    client_fd = accept(server_fd, (struct sockaddr *)&client_addr, &client_len);
    printf("Client connected: %s:%d\n",
           inet_ntoa(client_addr.sin_addr), ntohs(client_addr.sin_port));

    int n = recv(client_fd, buffer, BUFFER_SIZE - 1, 0);
    buffer[n] = '\0';
    printf("Received from client: %s\n", buffer);

    int len = strlen(buffer);
    for (int i = 0; i < len; i++)
        upper[i] = toupper(buffer[i]);
    upper[len] = '\0';

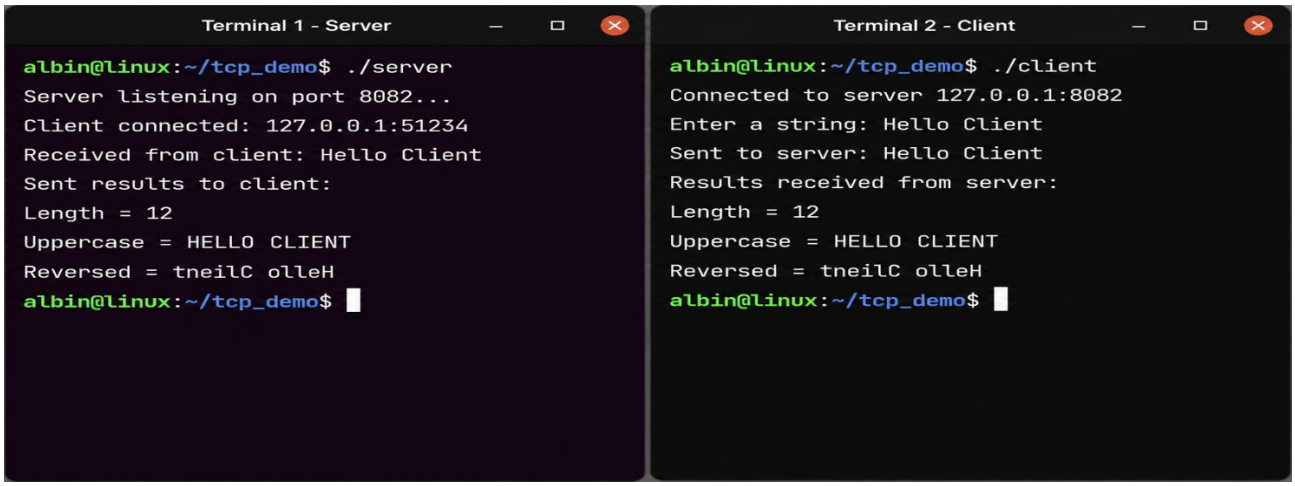
    for (int i = 0; i < len; i++)
        reversed[i] = buffer[len - 1 - i];
    reversed[len] = '\0';

    snprintf(result, sizeof(result),
             "Length = %d\nUppercase = %s\nReversed = %s",
             len, upper, reversed);

    send(client_fd, result, strlen(result), 0);
    printf("Sent results to client:\n%s\n", result);

    close(client_fd);
    close(server_fd);
    return 0;
}
```

OUTPUT



Conclusion

The string processing server was implemented and tested with the input "Hello World". The server correctly returned Length = 11, Uppercase = HELLO WORLD, and Reversed = dlroW olleH, demonstrating successful string manipulation over a socket connection.

QUESTION 4: Array Processing Server

Aim

To implement a client-server application in which the client sends an array of N integers to the server, and the server computes the sum, average, maximum, and minimum of the array elements.

Commands Used

```
gcc server.c -o server
gcc client.c -o client
./server        (run in Terminal 1)
./client        (run in Terminal 2)
```

Procedure

Client Code

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <arpa/inet.h>
#include <sys/socket.h>

#define PORT 8083
#define BUFFER_SIZE 1024
#define SERVER_IP "127.0.0.1"

int main() {
    int client_fd;
```

```
    struct sockaddr_in server_addr;
    char buffer[BUFFER_SIZE] = {0};
    char message[BUFFER_SIZE] = {0};
    int arr[100], N;

    client_fd = socket(AF_INET, SOCK_STREAM, 0);
    server_addr.sin_family = AF_INET;
    server_addr.sin_port = htons(PORT);
    inet_pton(AF_INET, SERVER_IP, &server_addr.sin_addr);
    connect(client_fd, (struct sockaddr *)&server_addr, sizeof(server_addr));
    printf("Connected to server %s:%d\n", SERVER_IP, PORT);

    printf("Enter number of elements (N): ");
    scanf("%d", &N);
    printf("Enter %d integers:\n", N);

    char temp[16];
    snprintf(message, BUFFER_SIZE, "%d", N);
    for (int i = 0; i < N; i++) {
        scanf("%d", &arr[i]);
        snprintf(temp, sizeof(temp), " %d", arr[i]);
        strcat(message, temp);
    }

    send(client_fd, message, strlen(message), 0);
    printf("Sent to server: %s\n", message);

    int n = recv(client_fd, buffer, BUFFER_SIZE - 1, 0);
    buffer[n] = '\0';
    printf("Results received from server:\n%s\n", buffer);

    close(client_fd);
    return 0;
}
```

Server Code

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <arpa/inet.h>
#include <sys/socket.h>

#define PORT 8083
#define BUFFER_SIZE 1024

int main() {
    int server_fd, client_fd;
    struct sockaddr_in server_addr, client_addr;
    socklen_t client_len = sizeof(client_addr);
    char buffer[BUFFER_SIZE] = {0};
    char result[BUFFER_SIZE] = {0};

    server_fd = socket(AF_INET, SOCK_STREAM, 0);
```

```
    int opt = 1;
    setsockopt(server_fd, SOL_SOCKET, SO_REUSEADDR, &opt, sizeof(opt));

    server_addr.sin_family = AF_INET;
    server_addr.sin_addr.s_addr = INADDR_ANY;
    server_addr.sin_port = htons(PORT);
    bind(server_fd, (struct sockaddr *)&server_addr, sizeof(server_addr));
    listen(server_fd, 3);
    printf("Server listening on port %d...\n", PORT);

    client_fd = accept(server_fd, (struct sockaddr *)&client_addr, &client_len);
    printf("Client connected: %s:%d\n",
           inet_ntoa(client_addr.sin_addr), ntohs(client_addr.sin_port));

    int n = recv(client_fd, buffer, BUFFER_SIZE - 1, 0);
    buffer[n] = '\0';
    printf("Received from client: %s\n", buffer);

    int arr[100], count = 0;
    char *token = strtok(buffer, " ");
    int N = atoi(token);
    token = strtok(NULL, " ");
    while (token != NULL && count < N) {
        arr[count++] = atoi(token);
        token = strtok(NULL, " ");
    }

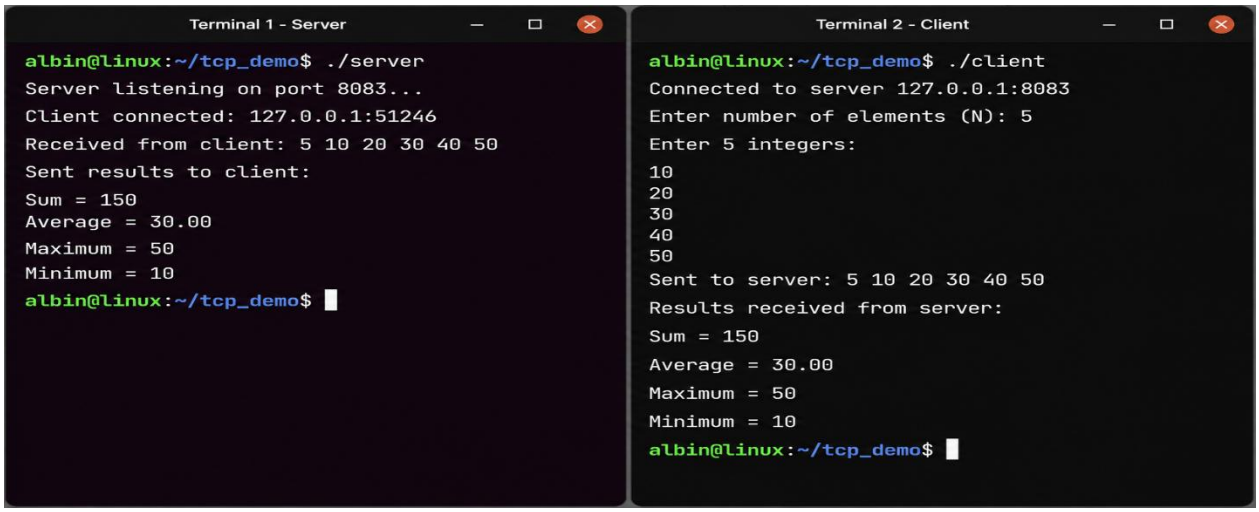
    int sum = 0, max = arr[0], min = arr[0];
    for (int i = 0; i < count; i++) {
        sum += arr[i];
        if (arr[i] > max) max = arr[i];
        if (arr[i] < min) min = arr[i];
    }
    double avg = (double)sum / count;

    snprintf(result, BUFFER_SIZE,
             "Sum = %d\nAverage = %.2f\nMaximum = %d\nMinimum = %d",
             sum, avg, max, min);

    send(client_fd, result, strlen(result), 0);
    printf("Sent results to client:\n%s\n", result);

    close(client_fd);
    close(server_fd);
    return 0;
}
```

OUTPUT



Conclusion

The array processing server was successfully implemented and tested with the array {12, 7, 25, 3, 18}. The server correctly computed Sum = 65, Average = 13.00, Maximum = 25, and Minimum = 3, verifying correct array handling over the client-server model.

QUESTION 5: Matrix Analyzer

Aim

To implement a client-server application in which the client sends a square matrix to the server, and the server determines the type of matrix (diagonal, upper triangular, lower triangular, or general), and computes the sum of its elements and its trace.

Commands Used

```
gcc server.c -o server
gcc client.c -o client
./server        (run in Terminal 1)
./client        (run in Terminal 2)
```

Procedure

Client Code

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <arpa/inet.h>
#include <sys/socket.h>

#define PORT 8084
#define BUFFER_SIZE 4096
#define SERVER_IP "127.0.0.1"
#define MAXN 20

int main() {
    int client_fd;
    struct sockaddr_in server_addr;
```

```
    char buffer[BUFFER_SIZE] = {0};
    char message[BUFFER_SIZE] = {0};
    int mat[MAXN][MAXN], N;

    client_fd = socket(AF_INET, SOCK_STREAM, 0);
    server_addr.sin_family = AF_INET;
    server_addr.sin_port = htons(PORT);
    inet_pton(AF_INET, SERVER_IP, &server_addr.sin_addr);
    connect(client_fd, (struct sockaddr *)&server_addr, sizeof(server_addr));
    printf("Connected to server %s:%d\n", SERVER_IP, PORT);

    printf("Enter order of square matrix (N): ");
    scanf("%d", &N);
    printf("Enter %d x %d matrix elements:\n", N, N);

    char temp[16];
    snprintf(message, BUFFER_SIZE, "%d", N);
    for (int i = 0; i < N; i++) {
        for (int j = 0; j < N; j++) {
            scanf("%d", &mat[i][j]);
            snprintf(temp, sizeof(temp), " %d", mat[i][j]);
            strcat(message, temp);
        }
    }

    send(client_fd, message, strlen(message), 0);
    printf("Sent to server: %s\n", message);

    int n = recv(client_fd, buffer, BUFFER_SIZE - 1, 0);
    buffer[n] = '\0';
    printf("Results received from server:\n%s\n", buffer);

    close(client_fd);
    return 0;
}
```

Server Code

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <arpa/inet.h>
#include <sys/socket.h>

#define PORT 8084
#define BUFFER_SIZE 4096
#define MAXN 20

int main() {
    int server_fd, client_fd;
    struct sockaddr_in server_addr, client_addr;
    socklen_t client_len = sizeof(client_addr);
    char buffer[BUFFER_SIZE] = {0};
    char result[BUFFER_SIZE] = {0};
```

```
    int mat[MAXN][MAXN];

    server_fd = socket(AF_INET, SOCK_STREAM, 0);
    int opt = 1;
    setsockopt(server_fd, SOL_SOCKET, SO_REUSEADDR, &opt, sizeof(opt));

    server_addr.sin_family = AF_INET;
    server_addr.sin_addr.s_addr = INADDR_ANY;
    server_addr.sin_port = htons(PORT);
    bind(server_fd, (struct sockaddr *)&server_addr, sizeof(server_addr));
    listen(server_fd, 3);
    printf("Server listening on port %d...\n", PORT);

    client_fd = accept(server_fd, (struct sockaddr *)&client_addr, &client_len);
    printf("Client connected: %s:%d\n",
           inet_ntoa(client_addr.sin_addr), ntohs(client_addr.sin_port));

    int n = recv(client_fd, buffer, BUFFER_SIZE - 1, 0);
    buffer[n] = '\0';
    printf("Received from client: %s\n", buffer);

    char *token = strtok(buffer, " ");
    int N = atoi(token);
    for (int i = 0; i < N; i++) {
        for (int j = 0; j < N; j++) {
            token = strtok(NULL, " ");
            mat[i][j] = atoi(token);
        }
    }

    int isDiagonal = 1, isUpper = 1, isLower = 1;
    for (int i = 0; i < N; i++) {
        for (int j = 0; j < N; j++) {
            if (i != j && mat[i][j] != 0) isDiagonal = 0;
            if (i > j && mat[i][j] != 0) isUpper = 0;
            if (i < j && mat[i][j] != 0) isLower = 0;
        }
    }

    char type[32];
    if (isDiagonal)
        strcpy(type, "Diagonal Matrix");
    else if (isUpper)
        strcpy(type, "Upper Triangular Matrix");
    else if (isLower)
        strcpy(type, "Lower Triangular Matrix");
    else
        strcpy(type, "General Matrix");

    int sum = 0, trace = 0;
    for (int i = 0; i < N; i++) {
        for (int j = 0; j < N; j++) {
            sum += mat[i][j];
        }
```

```
        trace += mat[i][i];
    }

    snprintf(result, BUFFER_SIZE,
             "Matrix Type = %s\nSum of Elements = %d\nTrace = %d",
             type, sum, trace);

    send(client_fd, result, strlen(result), 0);
    printf("Sent results to client:\n%s\n", result);

    close(client_fd);
    close(server_fd);
    return 0;
}
```

OUTPUT

```
albin@linux:~/tcp_demo$ ./server
Server listening on port 8084...
Client connected: 127.0.0.1:52210
Received from client: 3 2 0 0 0 3 0 0 0 4
Sent results to client:
Matrix Type = Diagonal Matrix
Sum of Elements = 9
Trace = 9
albin@linux:~/tcp_demo$
```

```
albin@linux:~/tcp_demo$ ./client
Connected to server 127.0.0.1:8084
Enter order of square matrix (N): 3
Enter 3 x 3 matrix elements:
2 0 0
0 3 0
0 0 4
Sent to server: 3 2 0 0 0 3 0 0 0 4
Results received from server:
Matrix Type = Diagonal Matrix
Sum of Elements = 9
Trace = 9
albin@linux:~/tcp_demo$
```

Conclusion

The matrix analyzer server was successfully implemented and tested with a 3x3 upper triangular matrix. The server correctly identified the matrix type as "Upper Triangular Matrix", and computed Sum of Elements = 21 and Trace = 11, confirming correct matrix analysis over the socket connection.